

Piyush Sharma

New Delhi, Delhi

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Full-Stack Developer | Backend Engineer | AI Applications Engineer

Education

Thapar Institute of Engineering and Technology — BTech in Computer Science

Oct 2022 – Jun 2026

Experience

Software Engineer Intern | Starlight Tech Services Pvt Ltd

Mar 2026 – Present

- Built backend services and async workflows for a full-stack PERN application across mobile and desktop platforms, implementing REST APIs, relational database schemas, and media-processing features.
- Developed a Python-based image optimization pipeline converting images to WebP with aspect-ratio preservation, reducing sizes by 60–80%; set up CI/CD deployment workflows for reliable production releases.

Software Engineer Intern | Synexoo

Jan 2025 – Mar 2025

- Developed backend services for a social media platform using Node.js, Express.js, and MongoDB, implementing user profiles, follow systems, and optimized database queries.
- Built a real-time messaging system using Socket.io with stable event handling and bidirectional communication for concurrent sessions.

Projects

Creator Intelligence | Node.js, Express.js, Supabase, Groq, Gemini, OpenRouter, Serper

[GitHub](#) [Live](#)

- Built an LLM-powered creator intelligence platform leveraging Groq, Gemini, and OpenRouter to generate market research reports, identify content gaps, and surface emerging opportunities before market saturation.
- Designed AI reasoning and retrieval pipelines for opportunity, interest, and competition scoring using real-time web retrieval (Serper), multi-provider model fallback, prompt orchestration, and structured report generation; integrated Google OAuth, Supabase authentication, and persistent analysis history.

MediaFlux | Flutter (Android/iOS), FastAPI, Python

[GitHub](#) [APK](#)

- Built a cross-platform media optimization app using Flutter and FastAPI supporting batch uploads, AVIF/WEBP conversion, resizing workflows, and real-time tracking; designed async job-processing with threaded background tasks and progressive result rendering.
- Developed media analysis pipelines generating heatmaps, compression metrics, and quality insights; implemented workspace isolation, temp file cleanup workers, and deployment workflows for large-batch processing stability.

SentinelAI | NestJS, Redis, Docker, GitHub Actions

[GitHub](#)

- Built an API security gateway with adaptive risk scoring to dynamically allow, throttle, or block suspicious requests; implemented Redis-based sliding window rate limiting to detect abnormal bursts during stress testing.
- Developed admin monitoring APIs for traffic inspection and manual IP controls; containerized via Docker with GitHub Actions CI/CD on Railway for reliable production deployment.

VCMtalker AI Assistant | Python, Ollama, Gemma, PyInstaller (Windows/macOS)

[GitHub](#) [Download](#)

- Developed a fully offline AI assistant using Ollama and Gemma for natural-language interaction, local LLM inference, voice-command execution, and OS automation; engineered a voice-processing pipeline supporting 50+ actions including application control, file operations, web workflows, and real-time system monitoring while maintaining complete local-data privacy.
- Built an intent-classification and action-routing engine that translated natural-language commands into executable workflows, enabling context-aware task execution, application launching, file management, browser automation, and system-control operations with low-latency local inference.

Skills

Programming: Python, C++, Java, JavaScript, TypeScript, SQL, Dart, C#

AI & Machine Learning: Prompt Engineering, NLP, AI Agents, Ollama, Gemma, Groq, Gemini, OpenRouter, Multi-LLM Systems, AI Workflow Automation, Structured Output Generation, Model Routing, Retrieval-Augmented Workflows, Local LLM Deployment

Backend & Databases: FastAPI, Node.js, Express.js, NestJS, REST APIs, WebSockets, Redis, MongoDB, MySQL, SQLite, Authentication, Rate Limiting, API Security

Cloud & DevOps: Docker, GitHub Actions, CI/CD, AWS, Linux, Git, Vercel, Railway, Supabase

Computer Science Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, Concurrency

Frontend & Mobile: Flutter, React.js, Next.js, Cross-platform Development, Responsive UI

Competitive Programming

- LeetCode** — Rating 1650+ (top 17.3%), solved 600+ problems across arrays, graphs, dynamic programming, greedy algorithms, and trees.
- CodeChef** (3-star) — Rating 1600+, solved 200+ problems.